FIITJEE

ALL INDIA TEST SERIES

FULL TEST – I

JEE (Main)-2023

TEST DATE: 28-12-2022

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 questions.

  Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Physics

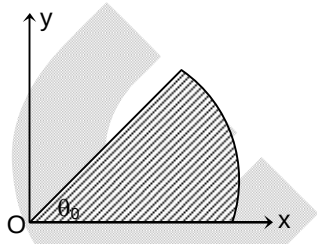
PART – A

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

1. Co-ordinates of the centre of mass of a sector of uniform circular disc of radius R and of mass m and subtends an angle θ_0 at it's centre as shown in figure are



(A) $\left[\frac{2R}{3}\frac{\sin\frac{\theta_0}{2}}{\frac{\theta_0}{2}}, \frac{2R}{3}\left(\frac{1-\cos\frac{\theta_0}{2}}{\frac{\theta_0}{2}}\right), 0\right]$

(B) $\left[\frac{2R}{3}\frac{\sin\theta_0}{\theta_0}, \frac{2R}{3}\left(\frac{1-\cos\theta_0}{\theta_0}\right), 0\right]$

(C) $\left[\frac{4R}{3}\frac{\sin\theta_0}{\theta_0}, \frac{4R}{3}\frac{\cos\theta_0}{\theta_0}, 0\right]$

(D) $\left[\frac{4R}{3}\frac{\sin\theta_0}{\theta_0}, \frac{2R}{3}\left(\frac{1-\cos\theta_0}{\theta_0}\right), 0\right]$

2. Two blocks A & B of mass 'm' & 2m respectively are joined to the ends of an under formed massless spring of spring constant 'k'. They can move on a horizontal smooth surface. Initially A & B has velocities 'u' towards left and '2u' towards right respectively. Constant forces of magnitudes F and 2 F are always acting on A and B respectively in the directions shown. The maximum extension in the spring during the motion is



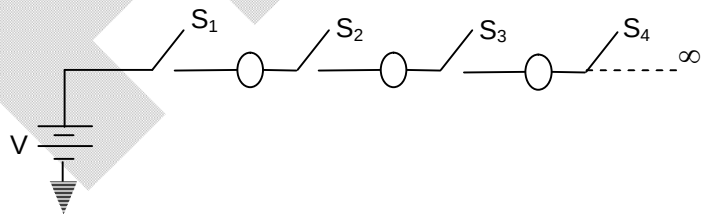
(A) $x_{max} = \frac{4F - \sqrt{16F^2 + 54mu^2k}}{3k}$

(B) $x_{max} = \frac{4F + \sqrt{16F^2 + 54mu^2k}}{3k}$

(C) $x_{max} = \frac{4F + \sqrt{16F^2 - 54mu^2k}}{3k}$

(D) $x_{max} = \frac{4F - \sqrt{16F^2 - 54mu^2k}}{3k}$

3. All spheres are identical of radius R and initially uncharged, S_1 is closed at t = 0, leaving all other switches open. After sufficiently long time S_1 opened and S_2 is closed. The process is done till S_{n-1} is opened and S_n is closed. Net loss of energy of system after t = 0 if process is done for infinite number of spheres, is



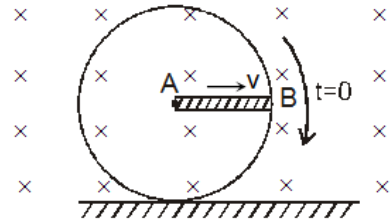
(A) $\frac{20}{6}\pi\varepsilon_0 RV^2$

(B) $2\pi\varepsilon_0 RV^2$

(C) $\frac{8}{3}\pi\varepsilon_0 RV^2$

(D) Zero

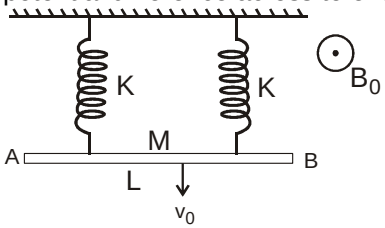
4. As shown in figure a conducting rod AB is attached to wooden disc of radius R and disc is in pure rolling with linear velocity v, find net emf developed across rod of length r as a function of time. Initial position of rod is shown in figure. Magnetic field is B and is directed into the plane.

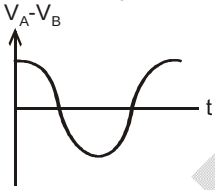


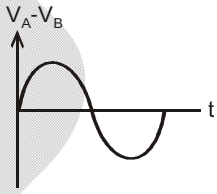
(A) $2BvR\left(1+\sin\frac{v}{R}t\right)$ (B) zero

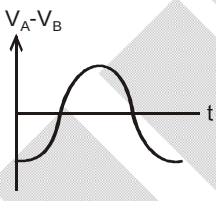
(C) $\frac{BvR}{2}\left(1+2\cos\frac{v}{R}t\right)$ (D) $\frac{BvR}{2}\left(1-2\sin\frac{v}{R}t\right)$

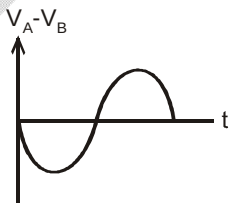
5. A horizontal conducting rod is kept in vertical plane is supported by two non-conducting springs as shown. A uniform magnetic field is directed outward rod initially at rest in equilibrium position and t=0 it is given velocity v_0 in downward direction. Which of the following graphs represent potential difference across to ends of the rod.



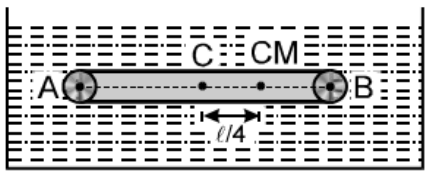
(A) 

(B) 

(C) 

(D) 

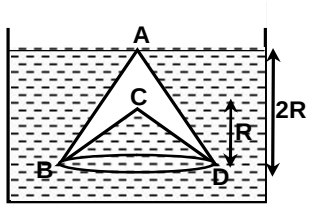
6. A non uniform cylinder of mass m, length ℓ and radius r is having its centre of mass at a distance ℓ/4 from the centre and lying on the axis of the cylinder. The cylinder is kept in a liquid of uniform density ρ. The moment of inertia of the rod about the centre of mass is I. The angular acceleration of point A relative to point B just after the rod is released from the position shown in figure is



(A) $\frac{\pi\rho g\ell^2r^2}{I}$ (B) $\frac{\pi\rho g\ell^2r^2}{4I}$

(C) $\frac{\pi\rho g\ell^2r^2}{2I}$ (D) $\frac{3\pi\rho g\ell^2r^2}{4I}$

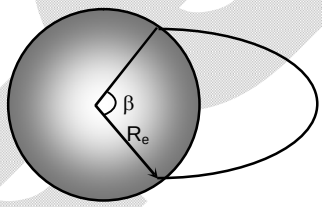
7. A solid cone of uniform density and height 2R and the base radius R has a conical portion scooped out from its base with the same base radius but height R as shown in the figure. The solid cone is floating in a liquid of density r with vertex A touching the fluids surface. If atmospheric pressure is P_0, the force acting on the surface BCD due to liquid is



(A) $\frac{\pi R^3 \rho g}{3}\left(2+\frac{2P_0}{R\rho g}\right)$ (B) $\frac{\pi R^3 \rho g}{3}\left[5+\frac{3P_0}{R\rho g}\right]$

(C) $\frac{\pi R^3 \rho g}{3}\left[1+\frac{3P_0}{R\rho g}\right]$ (D) $\frac{\pi R^3 \rho g}{3}\left[4+\frac{3P_0}{R\rho g}\right]$

8. A particle is projected from the surface of the earth (of radius R_e and mass M_e) with speed equal to $\sqrt{\frac{GM_e}{R_e}}$, at certain angle from local horizontal as shown in the figure such that the angle subtended by arc between launching & landing site at earth's centre is $\beta = 90°$. Calculate the maximum separation of the particle from the centre of the earth. Consider earth to be uniformly dense and air resistance to be absent.

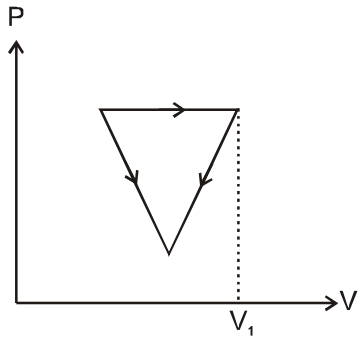


(A) $R_e\sqrt{2}$ (B) $R_e(\sqrt{2}+1)$

(C) $R_e\frac{\left(\sqrt{2}+1\right)}{\sqrt{2}}$ (D) $R_e\frac{\left(\sqrt{2}+1\right)}{\left(\sqrt{2}-1\right)}$

9. A satellite is moved from one circular orbit around the earth, to another of lesser radius. Which of the following statement is true?
(A) The kinetic energy of satellite increases and the gravitational potential energy of satellite-earth system increases;
(B) The kinetic energy of satellite increases and the gravitational potential energy of satellite-earth system decreases;
(C) The kinetic energy of satellite decreases and the gravitational potential energy of satellite-earth system decreases;
(D) The kinetic energy of satellite decreases and the gravitational potential energy of satellite-earth system increases.

10. The diagram shows an equilateral cyclic process on a PV diagram. The ratio of maximum pressure to minimum pressure is 2. The maximum temperature in the process is T_1. The work done by the two moles of ideal gas in the cyclic process is equal to



(A) $\frac{R^2T_1^2}{\sqrt{3}V_1^2}$ (B) $\frac{R^2T_1^2}{V_1^2}$

(C) $\frac{3R^2T_1^2}{V_1^2}$ (D) $\frac{3}{2}\frac{R^2T_1^2}{V_1^2}$

11. The surface temp of a small planet having a circular orbit round the sun with time period T assuming sun and the planet to be black bodies. Take radius of sun = R mass of sun = M, surface temp of the sun Q_0 is :

(A) $\sqrt{\frac{R}{2r}}\theta_0$ (B) $\sqrt{\frac{R+r}{2}}\theta_0$

(C) $\sqrt{\frac{R-r}{2}}\theta_0$ (D) $\sqrt{\frac{R}{r}}\theta_0$

12. In a sample of rock, the ratio of ^{206}Pb to ^{238}U nuclei is found to be 0.5. The age of the rock is. (Assume that all the Pb nuclides in the rock was produced due to the decay of Uranium nuclides and $T_{1/2}(^{238}U) = 4.5\times10^9$ year).

(A) 2.25×10^9 year (B) $4.5 \times 10^9 \ln 3$ year

(C) $4.5 \times 10^9 \frac{\ln\left(\frac{3}{2}\right)}{\ln 2}$ year (D) $2.25 \times 10^9 \ln\left(\frac{3}{2}\right)$ year

13. A body of mass m_0 is placed on a smooth horizontal surface. The mass of the body is decreasing exponentially with disintegration constant λ. Assuming that the mass is ejected backwards with relative velocity u. initially the body was at rest. The velocity of it after time t :

(A) $u\lambda t$ (B) $u e^{-\lambda t}$

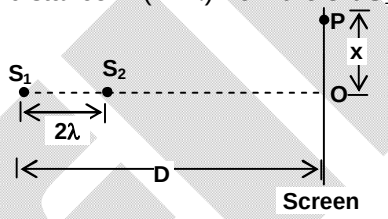
(C) $u e^{\lambda t}$ (D) none of these

14. A point object is moving with a velocity, $\vec{v}_0 = 2\hat{i} - 3\hat{j} + 4\hat{k}$; in front of a moving plane mirror whose normal is along x-axis. The mirror is moving with a velocity $\hat{i} - 4\hat{j} + 3\hat{k}$. The velocity vector of the image is given by

(A) $15\hat{j} + 3\hat{k}$ (B) $-3\hat{j} + 4\hat{k}$

(C) $\hat{j} + 2\hat{k}$ (D) $2\hat{i} + 3\hat{j} - 2\hat{k}$

15. Two coherent narrow slits emitting light of wavelength λ in the same phase are placed parallel to each other at a small separation of 2λ. The light is collected on screen which is placed at a distance D (>> λ) from the slit S_1 as shown in figure.



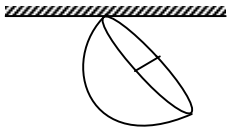
(A) The finite distance x such that the intensity at P is equal to the intensity at O is $\sqrt{3}D$

(B) The finite distance x such that the intensity at P is equal to the intensity at O is $\sqrt{\frac{3}{2}}D$

(C) There is no point on the screen at which the intensity at P is equal to intensity at O

(D) The finite distance x such that the intensity at P is equal to the intensity at O is $\sqrt{2}D$

16. Time period of small oscillations of a uniform hemispherical shell of radius R and mass m is



(A) $2\pi\sqrt{\frac{2\sqrt{5}R}{3g}}$ (B) $2\pi\sqrt{\frac{2\sqrt{5}R}{g}}$

(C) $2\pi\sqrt{\frac{\sqrt{5}R}{2g}}$ (D) $2\pi\sqrt{\frac{\sqrt{5}R}{3g}}$

17. A siren is placed at a railway platform is emitting sound of frequency 10 KHz. A passenger sitting in a moving train A records frequency of 11 kHz, while the train approaches the siren. During his return journey in a different train B he records a frequency of 12 KHz while approaching the same siren. Ratio of velocity of train B to that of train A is

(A) 2 (B) $\frac{242}{252}$

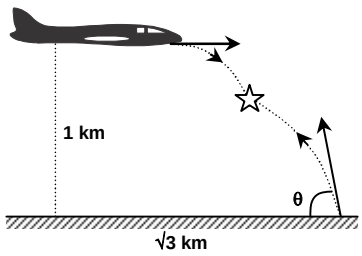
(C) $\frac{6}{5}$ (D) $\frac{11}{6}$

18. The pressure of an ideal gas varies according to the law $P = P_0 - AV^2$ where P_0 and A are positive constants. The highest temperature that can be attained by the gas is

(A) $\frac{P_0}{nR}\sqrt{\frac{P_0}{3A}}$ (B) $\frac{2P_0}{nR}\sqrt{\frac{P_0}{3A}}$

(C) $\frac{2P_0}{3nR}\sqrt{\frac{P_0}{3A}}$ (D) $\frac{2P_0}{3nR}\sqrt{\frac{P_0}{A}}$

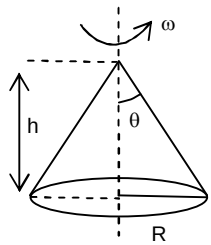
19. An aircraft is flying horizontally with a constant velocity = 200 m/s, at a height = 1 km above the ground. At the moment shown, a bomb is released from the aircraft and the cannon-gun below fires a shell with initial speed = 200 m/s, at some angle θ. For what value of 'θ' will the projectile shell destroy the bomb in mid air?



(A) $\theta = 15°$ (B) $\theta = 30°$

(C) $\theta = 45°$ (D) $\theta = 60°$

20. 'Q' charge is uniformly distributed over the surface of a right circular cone of semi-vertical angle θ and height h. The cone is uniformly rotated about its axis at angular velocity ω. Magnetic moment associated with the cone is found to be



(A) $\frac{Q\omega h^2 \tan^2\theta}{2}$ (B) $Q\omega h^2 \tan^2\theta$

(C) $\frac{Q\omega h^2 \tan^2\theta}{4}$ (D) $\frac{Q\omega h \tan\theta}{2}$

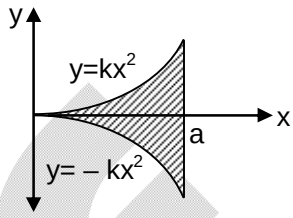
SECTION – B

(Numerical Answer Type)

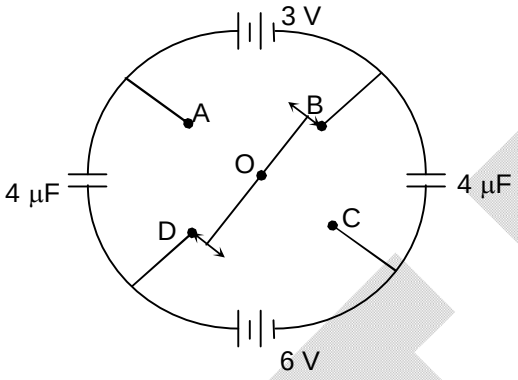
This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

21. A thin sheet of metal of uniform thickness is cut into the shape bounded by the line x = a and $y = \pm kx^2$, as shown.

If the x-coordinate of the centre of mass is $\frac{na}{12}$ then find n.



22. In figure there is four way key at the middle. If key is thrown from situation BD to AC, then charge flowing through point is q_0. Value of $\frac{q_0}{12}$ is ……

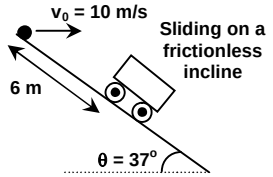


23. A fixed container of height 'H' with large cross-sectional area 'A' is completely filled with water. Two small orifices of cross-sectional area 'a' are made, one at the bottom and the other on the vertical side of the container at a distance H/2 from the top of the container. The time taken by the water level to reach a height of H/2 from the bottom of the container is $\frac{XA}{3\alpha}\left(\sqrt{2}-1\right)\sqrt{\frac{H}{g}}$. Find the value of X.

24. A wire forming a loop is dipped into soap solution and taken out, so that a film of soap solution is formed. A loop of 6.28 cm long thread is gently put on the film is pricked with a needle inside the loop. The thread loop takes the shape of a circle. Find the tension in the thread in (10^{-4} N). Surface tension of soap solution = 0.03 N/m.

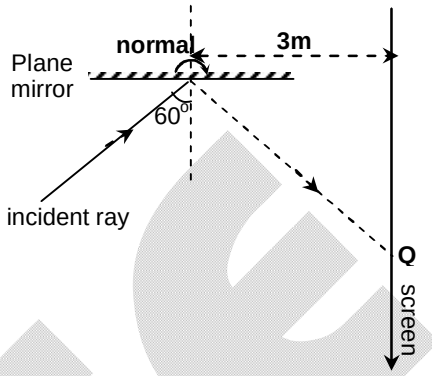
25. A solid copper sphere (density ρ and specific heat C) of radius r at an initial temperature 200 K is suspended inside a chamber whose walls are almost at 0 k. the time required for the temperature of sphere to drop to 100 K is found to be $\frac{x}{72}\frac{r\rho c}{\sigma}\times 10^{-6}$ sec. The value of *x* is.

26. A ball is projected horizontally from an incline so as to strike a cart sliding on the incline. Neglect height of cart and point of projection of ball above incline. At the instance the ball is thrown, the speed of cart is 'v' (in m/s). Find 'v' so that the ball strikes the cart.

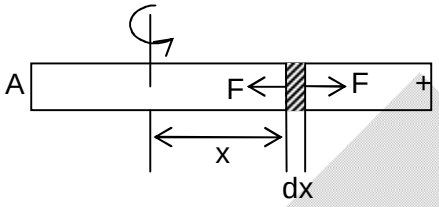


27. If the frequency of k_α x – ray emitted from the element with atomic number 31 is f, if the frequency of k_α X- ray emitted from the element with atomic number 51 is given by $\frac{25}{x}$ f then x will be ?

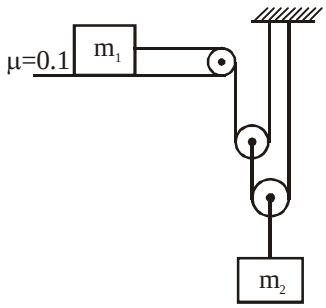
28. In the given figure, a ray is incident on the plane mirror as shown. The reflected ray strike the screen at Q. the mirror now starts, rotating about axis passing through P and perpendicular to plane of paper with angular speed $\pi/180$ rad/second. If the speed of point at which reflected ray strike the screen is $\frac{k\pi}{15}$ m/second at time t = 15 second, find the value of k.



29. A horizontally oriented copper rod of length l is rotated about a vertical axis passing through its middle. Frequency at which the rod ruptures when rotated at $\frac{1}{2\pi}\sqrt{\frac{X\sigma}{\rho l^2}}$. Breaking or breaking force of copper takes as σ and density of copper ρ. Find X.



30. In the figure, find the velocity of m_1 in ms^{-1} when m_2 falls by 9m.
Given $m_1 = m$
$m_2 = 2m$
(take $g = 10\ ms^{-2}$)



Chemistry

PART – B

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

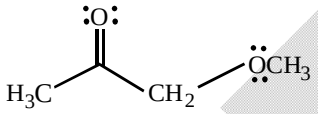
31. The conductivity measurement of a coordination compound of Cobalt (III) shows that it dissociates into 3 ions in solution. The compound is
(A) Hexaamminecobalt(III) chloride
(B) Pentaamminesulphatocobalt(III) chloride
(C) Pentaamminechloridocobalt(III) sulphate
(D) Pentaamminechloridocobalt(III) chloride

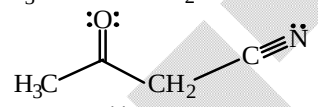
32. Which is the major product obtained by hydrolysis of compound formed by reaction between formaldehyde and ethyl magnesium bromide?
(A) Ethan-1-ol
(B) Propan-2-ol
(C) Propan-1-ol
(D) 2-Methyl-propan-2-ol

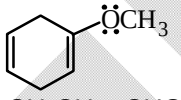
33. The total number of isomeric linear dipeptide which can be synthesized from racemic alanine is
(A) 1
(B) 2
(C) 3
(D) 4

34. The charge carried by 1 millimole of M^{n+} ions is 193 coulombs. The value of n is
(A) 1
(B) 2
(C) 3
(D) 4

35. The molecule/molecules that has/have delocalised lone pair(s) of electrons is/are

(I) 

(II) 

(III) 

(IV) $CH_3CH = CHCH_2CH_3$
(A) I, II and III
(B) I, II and IV
(C) I and III
(D) only III

36. Which one of the following electronic arrangements is absurd?
(A) n = 3, ℓ = 1, m = –1
(B) n = 3, ℓ = 0, m = 0
(C) n = 2, ℓ = 0, m = –1
(D) n = 2, ℓ = 1, m = 0

37. In order to oxidise a mixture one mole of each of FeC_2O_4, $Fe_2(C_2O_4)_3$, $FeSO_4$ and $Fe_2(SO_4)_3$ in acidic medium , the number of moles of $KMnO_4$ required is -
(A) 3
(B) 2
(C) 1
(D) 1.5

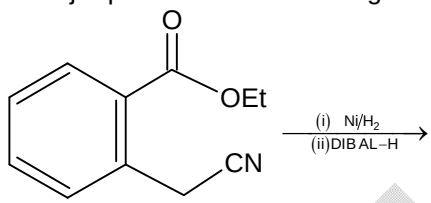
38. Which one of the following is likely to give a precipitate with $AgNO_3$ solution ?
(A) $(CH_3)_3CCl$
(B) $CHCl_3$
(C) $CH_2 = CH - Cl$
(D) CCl_4

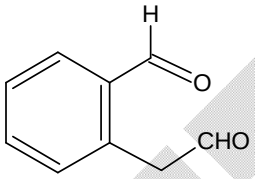
39. 1 g of non-volatile non-electrolyte solute is dissolved in 100g of two different solvents A and B whose ebullioscopic constants are in the ratio of 1 : 5. The ratio of the elevation in their boiling points, is :
(A) 5 : 1 (B) 10 : 1
(C) 1 : 5 (D) 1 : 0.2

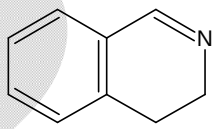
40. Among the following, the INCORRECT statement about colloids is :
(A) They can scatter light.
(B) They are large molecules and have high molar mass.
(C) The range of diameters of colloidal particles is between 1 and 1000 nm.
(D) The osmotic pressure of a colloidal solution is of higher order than the true solution for same weight of solute per litre solution.

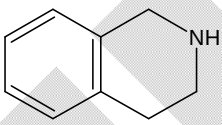
41. The radius of the largest sphere which fits properly at the centre of the edge of body centred cubic unit cell is :
(Edge length is represented by 'a') :-
(A) 0.134 a (B) 0.027 a
(C) 0.067 a (D) 0.047 a

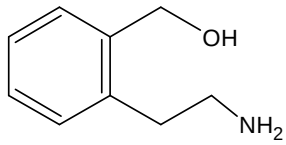
42. The major product of the following reaction is:



(A)


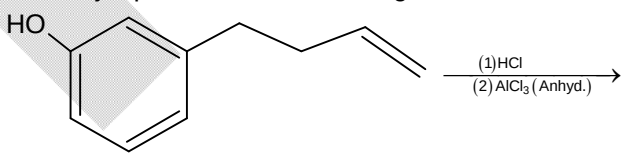
(B)


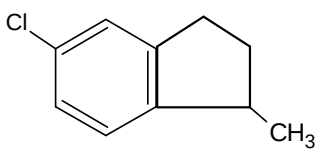
(C)


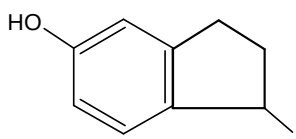
(D)


43. Three complexes, $[CoCl(NH_3)_5]^{2+}$ (I), $[Co(NH_3)_5H_2O]_3+$ (II) and $[Co(NH_3)_6]^{3+}$ (III)
absorb light in the visible region. The correct order of the wavelength of light absorbed by them is:
(A) (III) > (I) > (II) (B) (I) > (II) > (III)
(C) (II) > (I) > (III) (D) (III) > (II) > (I)

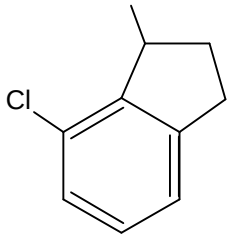
44. The major product of the following reaction is :



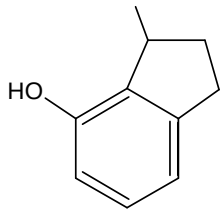
(A)


(B)


(C)



(D)



45. The primary pollutant that leads to photochemical smog is :
(A) Sulphur dioxide (B) Acrolein
(C) Ozone (D) Nitrogen oxides

46. NaH is an example of :
(A) Electron-rich hydride (B) Molecular hydride
(C) Saline hydride (D) Metallic hydride

47. For the reaction of H_2 with I_2, the rate constant is $2.5 \times 10^{-4} dm^3\ mol^{-1}\ s^{-1}$ at 327°C and 1.0 $dm^3\ mol^{-1}\ s^{-1}$ at 527°C. The activation energy for the reaction, in kJ mol^{-1} is:
($R = 8.314\ J\ K^{-1}\ mol^{-1}$)
(A) 72 (B) 166
(C) 150 (D) 59

48. Which of the following order is incorrect :
(A) Ionic character – $MCl > MCl_2 > MCl_3$
(B) Polarizability – $F^- < Cl^- < Br^- < I^-$
(C) Polarising power– $Na^+ < K^+ < Mg^{+2} < Al^{+3}$
(D) Polarising power – $Li^+ < Be^{+2} < B^{+3}$

49. $(CH_3)_2CHBr \xrightarrow[(ii)\ CuI]{(i)\ Li} A \xrightarrow{(CH_3)_2CHCH_2Br} B$

'B' is -
(A) $(CH_3)_2CHCH_2CH_3$
(B) $(CH_3)_2CHCH(CH_3)_2$
(C) $(CH_3)_2CHCH_2CH(CH_3)_2$
(D) $(CH_3)_2CHCH_2CH_2CH(CH_3)_2$

50. Calculate the potential of an indicator electrode versus the standard hydrogen electrode, which originally contained 0.1 M MnO_4^- and 1.72 M H^+ and which was treated with Fe^{2+} necessary to reduce 90% of $KMnO_4$ to Mn^{2+}

$E^0_{MnO_4^-/Mn^{2+}} = 1.51V$

(A) 1.4 V (B) 1.5 V
(C) 1.6 V (D) 1.3 V

SECTION – B

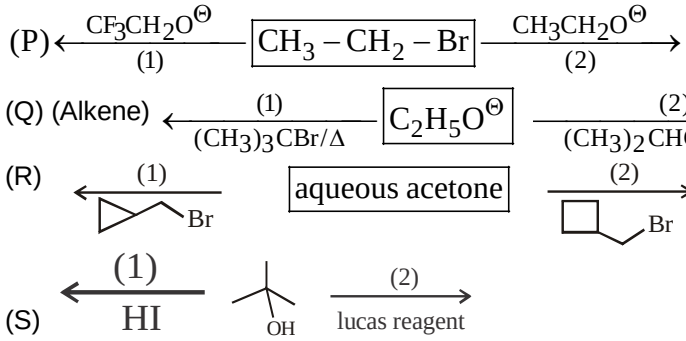
(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

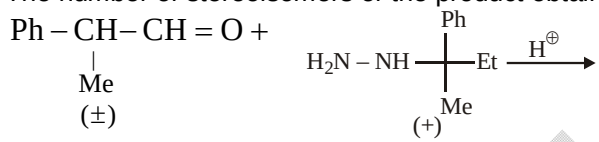
51. An adiabatic container fitted with a movable adiabatic piston (operating at 1 atm) is filled with 2 litre of 2M H_2O_2(aq) solution at 300 K. If H_2O_2 dissociates following first order decay with a half life of 10min then value of [Magnitude of change in internal energy (in cal) ÷ 100] in first 20 min will be

52. Number of compounds which are planar ?
XeF_2, ClF_3, H_2O, $[XeF_5]^-$, I_3^-, BCl_3, XeF_4, SF_4, PCl_5, SF_6, IF_7.

53. In how many reactions the 1st reaction is faster than 2nd reaction?



54. The number of stereoisomers of the product obtained in the following reaction is :



55. Na_2SO_3, NaCl, $Na_2C_2O_4$, Na_2HPO_4, Na_2CrO_4, $NaNO_2$, CH_3COONa are separately treated with $AgNO_3$ solution. In how many cases white ppt. is/are obtained?

56. A sample consisting of choclate-brown powder of PbO_2 is allowed to react with excess of KI and iodine liberated is reacted with N_2H_4 in another container. The volume of gas liberated from this second container at STP was measured out to be 1.12 litre. Find out volume of decimolar NaOH (in litre) required to dissolve PbO_2 completely. (Assume all reactions are 100% complete)

57. The ionization energy of the Be^{3+} ion with the electron in its ground state is 2.10×10^A kJ/mol. Find the value of A.

58. By how many times the absolute temperature of a gas when U_{rms} of a gas in a container of fixed volume is increased from 5×10^4 cm/sec to 10×10^4 cm/sec

59. A sample of HI(g) is placed in flask at a pressure of 0.2 atm. At equilibrium, the partial pressure of HI(g) is 0.04 atm. What is K_P for the given equilibrium ? $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$

60. In 100 ml sample of hard water, 100 ml of (N/50) Na_2CO_3 was added and the mixture was boiled and filtered. The filtrate was neutralized with 60 ml of (N/50) HCl. Calculate the permanent hardness of water (in ppm).
(sp. gr. of hard water = 1).

Mathematics

PART – C

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

61. A triangle is formed by the lines whose combined equation is given by $(x+y-4)(xy-2x-y+2)=0$ The equation of its circumcircle is

(A) $x^2+y^2-5x-3y+8=0$ (B) $x^2+y^2-3x-5y+8=0$

(C) $x^2+y^2-3x-5y-8=0$ (D) None of these

62. Two circles, each of radius 5, have a common tangent at (1,1) whose equation is 3x + 4y – 7 = 0. Then their centres are

(A) (4, -5), (-2, 3) (B) (4, -3), (-2, 5)

(C) (4, 5), (-2, -3) (D) None of these

63. If *p* and *q* be the longest distance and the shortest distance respectively of the point (-7, 2) from any point (α, β) on the curve whose equation is $x^2+y^2-10x-14y-51=0$, then G.M. of *p* and *q* is equal to

(A) $2\sqrt{11}$ (B) $5\sqrt{5}$

(C) 13 (D) None of these

64. Through the point (13, 31) a straight line is drawn to meet the axes of *X* and *Y* at Q and S respectively. If the rectangle OQRS is completed, the co-ordinates of R satisfy the equation.

(A) $\frac{13}{x}+\frac{31}{y}=1$ (B) $\frac{31}{x}+\frac{13}{y}=1$

(C) $\frac{13}{x}-\frac{31}{y}=1$ (D) $\frac{31}{x}-\frac{13}{y}=1$.

65. The normal chord of parabola, which sub-stends right angle at vertex is inclined at an angle α to the axis of parabola. Find the value of α ?

(A) $\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$ (B) $\tan^{-1}\left(\sqrt{7}\right)$

(C) $\frac{\pi}{3}$ (D) $\tan^{-1}\left(\sqrt{2}\right)$

66. A ray of light travelling along the line y = k from the positive direction of x-axis strikes a concave mirror, whose intersection with xy plane is parabola $y^2 = 4ax$. Find the equation of line along which the reflected ray will travel. Assume 'a' and 'k' are positive.

(A) $(k^2-4a^2)y=4ak(x-a)$ (B) $(k^2-4a^2)y=4ak(x+a)$

(C) $x=a$ (D) $x=2a$

67. TP and TQ are tangents drawn to a parabola from the point T and P_1, P_2 and P_3 are the lengths of perpendicular from P, T and Q respectively on any tangent to the curve. Then

(A) P_1, P_2, P_3 are in A.P (B) P_1, P_2, P_3 are in G.P

(C) P_1, P_2, P_3 are in H.P (D) $\frac{1}{P_1}+\frac{1}{P_2}=\frac{a}{P_2}$ (4a is latus rectum)

68. Any point P on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is joined to its centre O. The locus of mid-point of OP is

(A) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=\frac{1}{2}$ (B) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=\frac{1}{4}$

(C) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=2$ (D) $\frac{a^2}{x^2}+\frac{b^2}{y^2}=1$

69. If the normal at any point P on an ellipse of eccentricity 'e' meets the major axis and minor axis at T and G respectively, then PT : PG is equal to

(A) e (B) 1 : e

(C) $(1 - e^2) : 1$ (D) $(1 - e^2) : e$

70. For a certain value of c, $\lim_{x\to-\infty}\left[(x^5+7x^4+2)^C - x\right]$ is finite & non zero. The value of c and the value of the limit is

(A) 1/5, 7/5 (B) 0, 1

(C) 1, 7/5 (D) none of these

71. If $\int(x^3-2x^2+5)e^{3x}dx = e^{3x}(Ax^3+Bx^2+Cx+D)$ then the statement which is incorrect is

(A) C + 3D = 5 (B) A + B + 2/3 = 0

(C) 3C + 2B = 0 (D) A + B + C = 0

72. Let $I_1 = \int_0^{\pi/2} e^{-x^2}\sin(x)dx$; $I_2 = \int_0^{\pi/2} e^{-x^2}dx$; $I_3 = \int_0^{\pi/2} e^{-x^2}(1+x)dx$

and consider the statements

I $I_1 < I_2$ **II** $I_2 < I_3$ **III** $I_1 = I_3$

Which of the following is(are) true?

(A) I only (B) II only

(C) Neither I nor II nor III (D) Both I and II

73. If the function f(x) = 2 tan x + (2a + 1)ln |sec x| + (a –2)x is increasing in $\left(0, \frac{\pi}{2}\right)$ then range of 'a' equal to

(A) $(-\infty, 0]$ (B) [0, 1]

(C) [0, 3] (D) $[0, \infty)$

74. The maximum value of $\sec^{-1}\left(\frac{7-5(x^2+3)}{2(x^2+2)}\right)$ is:

(A) $\frac{5\pi}{6}$ (B) $\frac{5\pi}{12}$

(C) $\frac{7\pi}{12}$ (D) $\frac{2\pi}{3}$

75. Suppose p, q, r ≠ 0 and system of equations

$(p+a)x+by+cz=0$

$ax+(q+b)y+cz=0$

$ax+by+(r+c)z=0$

has a non-trivial solution, then value of $\frac{a}{p}+\frac{b}{q}+\frac{c}{r}$ is

(A) -1 (B) 0

(C) 1 (D) 2

76. If the system of equations $ax + hy + g = 0$, $hx + by + f = 0$ and $ax^2 + 2hxy + by^2 + 2gx + 2fy + c + t = 0$ has a unique solution, then

(A) t = 1 (B) t = 0

(C) $t=\frac{abc+2fgh-af^2-bg^2-ch^2}{h^2-ab}$ (D) $t=\frac{1}{h^2-ab}$

77. Let $\vec{p}$ and $\vec{q}$ be the position vectors of P and Q respectively, with respect to O and $|\vec{p}\,|=p$, $|\vec{q}|=q$. The points R and S divide PQ internally and externally in the ratio 2 : 3 respectively. If $\overrightarrow{OR}$ and $\overrightarrow{OS}$ are perpendicular, then

(A) $9p^2 = 4q^2$ (B) $4p^2 = 9q^2$

(C) 9p = 4q (D) 4p = 9q

78. If $\vec{d}=\lambda(\vec{a}\times\vec{b})+\mu(\vec{b}\times\vec{c})+\gamma(\vec{c}\times\vec{a})$ and $[\vec{a},\vec{b},\vec{c}]=1$

Then $\lambda+\mu+\gamma$ is equal to

(A) $\vec{d}.(\vec{a}+\vec{b}+\vec{c})$ (B) $\vec{a}+\vec{b}+\vec{c}$

(C) 1 (D) none of these

79. Area of the region bounded by the curve $y=\frac{16-x^2}{4}$ and $y = \sec^{-1}[-\sin^2 x]$ (where [.] denotes the greatest integer function) is

(A) $\frac{1}{3}(4-\pi)^{3/2}$ (B) $8(4-\pi)^{3/2}$

(C) $\frac{8}{3}(4-\pi)^{3/2}$ (D) $\frac{8}{3}(4-\pi)^{1/2}$

80. A contest consists of predicting the results (win, draw or defeat) of 10 football matches. The probability that one entry contains at least 5 correct answers is

(A) $\frac{12585}{3^{10}}$ (B) $\frac{12385}{3^{10}}$

(C) $\frac{9385}{3^{10}}$ (D) none of these

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. If sinx + cosx + tanx + cot x + secx + cosecx = 7 and sin2x = a – b$\sqrt{7}$, where a, b are rational , then $\frac{a+b}{10}$ is

82. If tanx – $\tan^2 x$ = 1, then the value of $\tan^4 x - 2\tan^3 x - \tan^2 x + 2\tan x + 1$ is

83. The hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ passes through (4, 2) and length of latus rectum is $\frac{4}{3}$, the angle between asymptotes is θ then $\frac{6}{\pi}\theta$ is

84. Co-ordinates of points common to hyperbola $25x^2 - 9y^2 = 225$ and 25x + 12y = 45 is (x_1, y_1) then $3|(x_1 + y_1)|$ is

85. For x satisfying $\log_{1/\sqrt{5}} (6^{x+1} - 36^x) \geq -2$, $x \in (-\infty, a] \cup [\log_b c, \infty)$ then a + b + c is

86. S.D of a data is 6. When each observation is increased by 1, then the S.D of new data is

87. Let u = $\int_0^1 \frac{\ln(x+1)}{x^2+1} dx$ and v = $\int_0^{\pi/2} \ln(\sin 2x)\, dx$ then 4u + v =

88. Let f(x) = Max. (cos x, x, 2x –1) where $x \geq 0$. Then number of points of non-differentiability of f(x), is equal to

89. If a_1 is the value of a for which function f(x) = $x^2 + \frac{a}{x}$ has a local minimum at x = 2 and a_2 is the value of a for which f(x) has a point of inflection at x = 1, then $\left(\frac{a_1 + a_2}{3}\right)$ is equal to

90. If n positive integers are taken at random and multiplied together, then the probability that the last digit of the product is 2, 4, 6 or 8 is $\frac{a^n - b^n}{C^n}$ then a + b + c is equal to